

AP02 clearance and free-fraction assessment, recovery-controlled dialysis

Final Report

Study MV-2018-052 (LIMS M18/BIO/01223)

Client: Avelwick Pharma

Study lead: Dr M. Larkin, DMPK Study Director

Report date: April 16, 2018

1 Summary

40 compounds from the Avelwick aminopyrimidine regioisomer series 02 were tested for Avelwick Pharma in the Microsomal CL_{int}, fu,p by equilibrium dialysis. Testing started on March 12, 2018 and results were reported on April 16, 2018. All accepted runs met matrix acceptance criteria.

Table 1 Study summary

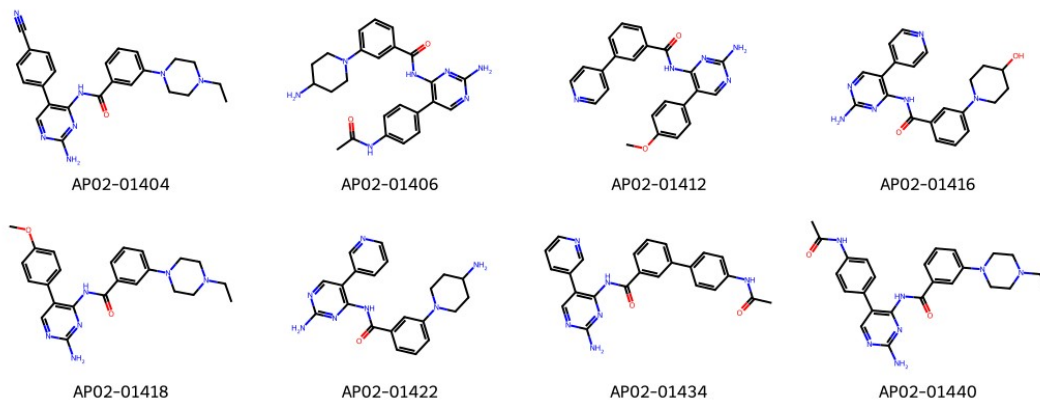
Metric	Value
Project	MV-2018-052
Client	Avelwick Pharma
Compounds tested	40
Series / chemotype	Avelwick aminopyrimidine regioisomer series 02 (Aminopyrimidine regioisomer)
Assays	Liver microsomal stability, Plasma protein binding (equilibrium dialysis)
Turnaround (calendar days)	35
Price	USD 23,500

2 Objective

To rank the test article set for Microsomal CL_{int}, fu,p by equilibrium dialysis liability so that Avelwick Pharma could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one accession lot on March 12, 2018 and logged against M18/BIO/01223. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



4 Methods

Microsomal CL_{int}

Test articles (1 μ M) were incubated with pooled human and rat liver microsomes (0.5 mg/mL protein) at 37 °C with NADPH (1 mM); a no-NADPH control at 45 minutes checks for non-metabolic loss. Samples at 0, 5, 15, 30 and 45 minutes are quenched in acetonitrile with internal standard and analyzed by LC-MS/MS. CL_{int} is calculated from the first-order depletion slope. When depletion over 45 minutes is below the quantifiable limit, CL_{int} is reported as an upper bound (<3 μ L/min/mg) and half-life as a lower bound (>60 min).

Matrix acceptance: verapamil (high clearance) and diltiazem (moderate clearance) within 0.5-2 times their historical CL_{int} in each run; no-NADPH loss below 15%.

f_{u,p} by equilibrium dialysis

Binding to human plasma was measured by rapid equilibrium dialysis (5 μ M test article, 4 h at 37 °C, PBS pH 7.4). Plasma and buffer compartments are matrix-matched before LC-MS/MS. Results are reported as percent bound and percent unbound with mass-balance recovery; recoveries below 70% are footnoted.

Warfarin is included as the binding control in every plate (acceptance: 97-99.5% bound).

5 Results

Microsomal stability

Table 2 Microsomal intrinsic clearance and half-life

Compound	Human CL _{int} (μ L/min/mg)	Human t _{1/2} (min)	Rat CL _{int} (μ L/min/mg)	Rat t _{1/2} (min)
AP02-01404	19	>60	47	29
AP02-01406	23	59	40	35
AP02-01412	21	>60	36	39
AP02-01416	18	>60	25	55
AP02-01418	58	24	113	12
AP02-01422	20	>60	36	39
AP02-01434	27	51	46	30
AP02-01440	18	>60	18	>60
AP02-01448	15	>60	32	43

AP02-01452	16	>60	38	37
AP02-01454	27	52	44	32
AP02-01460	21	>60	26	53
AP02-01464	25	55	48	29
AP02-01470	11	>60	15	>60
AP02-01476	11	>60	16	>60
AP02-01478	98	14	186	7.5
AP02-01488	23	>60	34	40
AP02-01494	18	>60	41	34
AP02-01496	11	>60	17	>60
AP02-01500	22	>60	37	38
AP02-01502	14	>60	46	30
AP02-01506	13	>60	11	>60
AP02-01508	4.1	>60	7.0	>60
AP02-01512	9.7	>60	13	>60
AP02-01518	62	22	91	15
AP02-01520	35	39	64	22
AP02-01526	35	39	53	26
AP02-01538	20	>60	28	49
AP02-01542	20	>60	36	39
AP02-01548	24	58	29	48
AP02-01550	21	>60	43	32
AP02-01556	81	17	113	12
AP02-01560	31	45	33	42
AP02-01566	12	>60	13	>60
AP02-01574	8.6	>60	20	>60
AP02-01578	34	41	65	21
AP02-01580	40	34	56	25
AP02-01584	20	>60	23	>60
AP02-01586	18	>60	31	45
AP02-01590	25	56	32	44

Controls: verapamil (high), diltiazem (medium) within historical range

Plasma protein binding

Table 3 Human plasma protein binding

Compound	% bound	Unbound (%)	Recovery (%)
AP02-01404	90.8	9.18	81
AP02-01406	90.3	9.73	99
AP02-01412	99.9	0.10	86
AP02-01416	79.6	20.43	82
AP02-01418	94.3	5.71	82
AP02-01422	81.6	18.44	88
AP02-01434	96.9	3.10	91
AP02-01440	93.0	7.04	102
AP02-01448	89.7	10.32	84
AP02-01452	97.2	2.80	82
AP02-01454	96.0	3.97	89
AP02-01460	80.1	19.85	88
AP02-01464	89.6	10.40	103
AP02-01470	89.2	10.84	94
AP02-01476	94.8	5.19	81
AP02-01478	99.9	0.10	98
AP02-01488	94.4	5.63	93
AP02-01494	99.5	0.50	93
AP02-01496	83.6	16.43	93

AP02-01500	93.0	6.96	83
AP02-01502	93.7	6.26	88
AP02-01506	87.5	12.46	96
AP02-01508	93.9	6.10	97
AP02-01512	88.8	11.22	99
AP02-01518	99.9	0.10	98
AP02-01520	89.8	10.24	90
AP02-01526	94.4	5.65	104
AP02-01538	92.3	7.66	102
AP02-01542	91.2	8.79	88
AP02-01548	99.0	1.04	83
AP02-01550	92.5	7.55	92
AP02-01556	98.7	1.35	100
AP02-01560	88.4	11.61	82
AP02-01566	89.8	10.18	101
AP02-01574	92.7	7.33	91
AP02-01578	99.7	0.33	90
AP02-01580	99.9	0.10	99
AP02-01584	87.6	12.42	96
AP02-01586	95.0	4.99	89
AP02-01590	99.8	0.22	88

Warfarin control 98-99% bound

6 Deviations

No deviations from the study plan.

7 Discussion

Microsomal clearance varied across the set and between species. Plasma binding should be considered with clearance when ranking exposure potential. Values are in vitro and do not by themselves predict clinical risk; free exposure estimates are needed for that assessment.

8 Conclusion

Recommended next steps: take the lower-clearance analogues into hepatocyte or in vivo follow-up.

9 Records

Raw data, assay worksheets and instrument files are retained for 5 years under M18/BIO/01223.

Name	Role	Signature	Date
Megan Larkin	DMPK Study Director	<u>M Larkin</u>	April 16, 2018
Helen Marchwood	Bioanalytical Quality Manager	<u>H Marchwood</u>	April 15, 2018